Taller: Derivadas Parciales

Polinomica:

$f(x,y) = x^4 + 3x^2y^3 - 5y^2$

$x^4 = 4x^3 \quad 3x^2y^3 = 6xy^3 \quad -5y^2 = 0$

$fx = 4x^3 + 6xy^3$ //

$x^4 = 0 \quad 3x^2y^3 = 9x^2y^2 \quad -5y^2 = -10y$

Exponencial:

$f(x,y) = e^{2x}\sin(y)$

$e^{2x} = 2e^{2x} \qquad fx = 2e^{2x}\sin(y)$ //

$\sin(y) \cdot \cos(y) \qquad fy = e^{2x}\cos(y)$ //

Logaritmica:

$f(x,y) = \ln(x^2 + y^2)$

$\frac{d}{dx}[\ln(u)] = \frac{u'}{u} \qquad u = x^2 + y^2 \qquad u' = 2x$

$fx = \frac{2x}{x^2 + y^2}$ // $\qquad u' = 2y \qquad fy = \frac{2y}{x^2 + y^2}$ //

Cociente:

$f(x,y) = \frac{x-y}{x+y} \qquad \frac{u}{v} = \frac{u'v - uv'}{v^2}$

$fx = \frac{(1)(x+y) - (x-y)(1)}{(x+y)^2} = \frac{x+y-x+y}{(x+y)^2}$

$$f_x = \frac{2y}{(x+y)^2} //$$

$$f_y = \frac{(-1)(x+y)-(x-y)(1)}{(x+y)^2} = \frac{-x-y-x+y}{(x+y)^2}$$

$$f_y = \frac{-2x}{(x+y)^2}$$

Tres Variables:

(f_x, f_y, f_z): $f(x,y,z) = x^2yz^3 + \cos(xz)$

- f_x

$x^2yz^3 = 2xyz^3$

$\cos(xz) = -\sin(xz)\cdot z$

$f_x = 2xyz^3 - z\sin(xz) //$

- f_y

$x^2yz^3 = x^2z^3$

$\cos(xz) = 0$

$f_x = x^2z^2 //$

- f_z

$x^2yz^3 = 3x^2yz^2$

$\cos(xz) = -\sin(xz)\cdot x$

$f_z = 3x^2yz^2 - x\sin(xz) //$

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